

Final Exam - Applied Abstract Algebra

June 9th, 2026

(40 points) Let Q be a quiver:

$$1 \longleftarrow 2 \longleftarrow 3 \longrightarrow 4 \longleftarrow 5$$

- a). Compute the Auslander-Reiten quiver for Q . Calculate $\text{Hom}(P(3), I(4))$.
- b). Construct a triangulation of polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(3)$, $I(4)$, $\tau(P(3))$, $\tau^{-1}(I(4))$ and to indecomposable representation of dimension vector $(1, 1, 1, 1, 1)$.

Choose up to three problems from this list to submit.

1. (20 points) Prove that P is projective if and only if $\text{Ext}^1(P, N) = 0$ for all representations N .
2. (20 points) Let $M \in \text{rep } Q$ and

$$0 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

be standard projective resolution. Show that

$$0 \rightarrow D(M) \rightarrow D(P_0) \rightarrow D(P_1) \rightarrow 0$$

is the injective resolution for $D(M) \in \text{rep } Q^{op}$. Show that duality functor D takes a minimal projective resolution for M to a injective envelope for $D(M)$.

3. (20 points) Let $g : M \rightarrow N$ be a surjective morphism between representations of Q , and let $P(i)$ be the projective representation at vertex i . Show that the map

$$g_* : \text{Hom}(P(i), M) \rightarrow \text{Hom}(P(i), N)$$

is surjective.

4. (20 points) Show that for any Q the representations $I(i)$, $i \in Q_0$ are indecomposable.
5. (20 points) Show that the smallest wild tree with valence at most 3 at each vertex has 7 vertices.